CT scan or computed tomography are special X-rays tests that produce cross-sectional images of the body using X-rays and a computer.

47. Which of the following was invented by Wilhelm Rontgen?

(a) Radio (b) X-Ray Machine
(c) Electric Bulb (d) None of the above

U.P. P.C.S. (Pre) 2018

Ans. (b)

X-ray, a type of electromagnetic waves was discovered by Wilhelm Rontgen, with wavelengths in the range of 0.01 to 10 nanometres. These rays are extensively used in the field of medical and industries.

Sound

Notes

Sound :

- Sound is that form of energy which produces a hearing sensation.
- Sound travels in the form of waves.
- Vibrating matters produce sound.
- The substance through which sound travels is called medium.
- Sound waves are mechanical waves.
- Medium (solid, liquid or gas) is necessary for their propagation.
- Sound is transmitted through gases, plasma, and liquids as longitudinal waves. Through solids, however, it can be transmitted as both longitudinal waves and transverse waves.

Speed of Sound :

- Sound speed depends upon the nature of the medium through which it propagates.
- Sound speed varies in different mediums.
- Sound propagates very slow in gas.
- Sound propagates faster than gas in a liquid.
- Sound speed is fastest in solids.
- Sound does not propagate in vacuum.
- Sound speed in steel is more than 15 times to speed of sound in air.
- Sound speed depends upon temperature.
- Increase in temperature results to increase in the speed of sound (in summer season the speed of sound is more than that of winter season).
- Humidity of air plays a very important role in the speed of sound.
- The sound speed in humid air is more than that of dry air.

Sounds speed in Different Mediums

S.No.	Medium	Sound Speed / Second
1.	Air (at 0ºC)	331 m/sec.
2.	Air (at 22ºC)	344 m/sec.
3.	Hydrogen (at 25ºC)	1284 m/sec.
4.	Seawater (at 25ºC)	1531 m/sec.
5.	Aluminium (at 25ºC)	6420 m/sec.
6.	Iron (at 25ºC)	5950 m/sec.
7.	Steel (at 25ºC)	5960 m/sec.

Frequency range of Sound :

- Sound waves have been classified into three types on the basis of their frequency range. These are as follows :

(i) Audible sound Waves :

- Our ear is only capable to hear such sound waves.
- Frequency range of such sound waves lies between 20 Hz to 20000 Hz.

(ii) Infrasonic Waves :

- Frequency of such sound waves is below the frequencies of audible sound i.e. below 20 hz.
- Sources of infrasound in nature include volcanoes, avalanches, earthquakes and meteorites.
- Many animals can hear infrasound like whales, elephants, rhinos, hippos, giraffes, alligators, squid/cuttlefish/octopus and even pigeons.
- Generally, it has been observed that, before the earthquake some animals become abnormal. Actually, earthquake produces infrasounds of low frequency before producing main shock waves, which probably alert the animals.

(iii) Ultrasonic Waves :

- Ultrasonic waves are such sound waves whose frequency is more than 20,000 Hz.
- Such sound waves are beyond the normal hearing range of human.
- A number of animals are capable of emitting ultrasonic frequencies and use it for several purposes such as - bats, whales, dolphins, mice, etc.

Applications of Ultrasonic Waves :

- A dog whistle **(Galton whistle)** is a whistle that emits ultrasound used for training and calling dogs.

- Ultrasonic waves are widely used in the field of industries and medicines.
- Image of the heart is obtained by reflecting the ultrasonic waves from different angle/part of the heart. This technique is known as Echocardiography (Echo Test).
- Ultrasound waves are used to breakdown the kidney stone into smaller pieces. These pieces are excreted through urine outside of the body.
- Ultrasound is used for cleaning a zig-zag tube.
- SONAR (Sound Navigation & Ranging) device is helpful in determining the distance of any object positioned in water and to measure the direction and speed of any object by using ultrasonic waves.

Sound Intensity :

- Sound intensity is defined as the power carried by sound waves per unit area in a direction perpendicular to that area.
- The SI Unit of intensity, which includes sound intensity, is watt/square metre (w/m^2).
- The unit of sound intensity is bel, but 1/10th part of bel is commonly used, which is termed as decibel (dB).

Echo :

- A sound or sounds caused by the reflection of sound waves from a surface back to the listener is known as an echo.
- The impulse of sound remains in existence for 0.1 second in mind. So, to hear clear echo there must be a time interval of at least 0.1 second between the original sound and reflected sound.
- At least 16.5 metre distance between listener and reflector is necessary to hear the clear echo.

Mach Number :

- It is the ratio of the speed of a body to the speed of sound in the surrounding medium.
- It is often used with numerals as Mach 1, Mach 2 etc.
- Mach Number = $\dfrac{\text{speed of object}}{\text{speed of sound in air}}$
- **Subsonic Aircraft :** An aircraft whose speed is less than the speed of sound.
- **Supersonic Objects :** The objects with Mach Number between 1-5, e.g. missiles.
- **Hypersonic Objects :** The objects whose Mach number is more than 5, e.g. BrahMos-II missile.

Question Bank

1. Put in ascending order of speed of sound in the mediums
I . Water, II . Steel, III . Nitrogen :

(a) III, II, I (b) III, I, II
(c) I, III, II (d) II, I, III

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (b)

The speed of sound is different for different type of mediums. The speed of sound is maximum in solids while minimum in gases. The speed of sound depends upon the density of the medium through which it is travelling. The medium which has higher density, the sound will travel faster in that medium.

2. The sound will have the highest velocity in :

(a) Vacuum (b) Air
(c) Water (d) Steel

U.P. P.C.S. (Pre) 2018

Ans. (d)

The speed of sound varies from medium to medium. Sound travels most slowly in gases, it travels faster in liquids and fastest in solids. For example - sound travels at 331 m/s in air, it travels at 1531 m/s in sea water and at 5950 m/s in iron.

3. The velocity of sound in air is approximately :

(a) 10 km./sec. (b) 10 mile/min.
(c) 330 m/sec. (d) 3×10^{10}/sec.

42nd B.P.S.C. (Pre) 1997

Ans. (c)

See the explanation of above question.

4. The velocity of sound is maximum in :

(a) Air (b) Liquid
(c) Metal (d) Vacuum

U.P. Lower (Spl) (Pre) 2008
Uttarakhand P.C.S. (Pre) 2010

Ans. (c)

See the explanation of above question.

5. If Va, Vw and Vs respectively are the speed of sound in air, water and steel, then :

(a) Va < Vw < Vs (b) Vs < Vw < Va
(c) Vw < Vs < Va (d) Vs < Va < Vw

U.P. U.D.A./L.D.A. (Spl) (Mains) 2010

Ans. (a)

As we know that the speed of sound is different for different types of medium. In general, sound travels faster in liquid than gases and faster in solid than in liquid.

6. **In which medium the speed of sound is maximum at a temperature of around 20°C ?**
(a) Air (b) Granite
(c) Water (d) Iron
Chhattisgarh P.C.S. (Pre) 2004
Ans. (d)

The speed of sound depends on the elasticity and density of the medium through which it is travelling. Greater the elasticity and the density, sound travels faster in that medium. At temperature of 20°C, the speed of sound is maximum in iron.

7. **Sound waves :**
(a) Can travel in vacuum.
(b) Can travel only in solid.
(c) Can travel only in gases.
(d) Can travel both in solid and gaseous medium.
U.P.P.C.S. (Pre) 2002
Ans. (d)

Sound waves need to travel through a medium such as a solid, liquid or gas. The sound waves travel through each of these mediums by vibrating the molecules in the matter. The molecules in solids are packed very tightly but in liquids are not packed as tightly as solid, and in gases they are very loosely packed. The spacing of the molecules, enable sound to travel much faster through solid than gases. So sound waves can travel in solid, liquid and gaseous medium.

8. **Sound waves travel fastest in :**
(a) solids (b) liquids
(c) gases (d) vacuum
M.P.P.C.S. (Pre) 2017
Ans. (a)

Sound is a mechanical wave and needs a material medium like air, water, steel etc. for its propagation. It cannot travel through vacuum. Speed of sound is different in different medium. Speed of sound is maximum in solid than liquid and gas.

9. **In which of the following option sound may not be across/travel?**
(a) Water (b) vacuum
(c) Iron (d) Air
U.P.P.C.S. (Pre) 1990
Ans. (b)

See the explanation of above question.

10. **Sound waves do not travel in :**
(a) Solids (b) Liquids
(c) Gases (d) Vacuum
U.P.P.C.S. (Pre) 2014
Ans. (d)

See the explanation of above question.

11. **The walls of the hall, built for music concerts should :**
(a) Amplify sound (b) Transmit sound
(c) Reflect sound (d) Absorb sound
U.P.P.C.S. (Mains) 2007
Ans. (d)

The walls of the hall built for music concerts should absorb sounds. Most of the solid walls reflects the sound. If the walls of concert hall reflect the sound, then audiences hear the echo sound. So to avoid this, there is a need to built soft surface walls.

12. **Which of the following is a cause of echo?**
(a) Absorption of sound (b) Transmission of sound
(c) Reflection of sound (d) Refraction of sound
Jharkhand P.C.S. (Pre) 2023
Ans. (c)

An echo is a sound caused by the reflection of sound waves from a surface back to the listener. It is the reflection of sound, arriving at the listener sometime after the direct sound. Echoes are a result of reflections when a sound wave comes into contact with a boundary, particularly a flat and hard surface. Basically a reflection of sound causes echo.

13. **To hear a clear echo, the minimum distance between the reflecting surface and the observer should be :**
(a) 165 feet (b) 165 metre
(c) 16.5 feet (d) 16.5 metre
U.P.P.C.S. (Mains) 2007
Ans. (d)

The repetition of sound produced due to the reflection from a large surface like wall, hill or mountain is called echo. Consider an observer is producing a sound and it gets reflected by an obstacle. The sound travel towards the observer and the observer hear the sound again. Let 'd' be the distance between the observer and the obstacle, 'v' is the sound velocity and 't' is the time taken by the sound to and fro motion, then the velocity of the sound is given by

$$v = 2d/t$$

Substituting $t = \frac{1}{10}$ sec, it is minimum time required to distinguish between two sounds

v = velocity of sound = 330 m/sec at 20° C

then $d = \frac{vt}{2} = \frac{1}{2}\left[330 \times (1/10)\right]$

$= 16.5$ m (about 17m)

14. **How much should minimum distance be between the source of sound and reflecting surface, so that an echo can be heard clearly?**
(a) 10 metre (b) 17 metre

(c) 24 metre (d) 30 metre

R.A.S./R.T.S.(Pre) 2007

Ans. (b)

See the explanation of above question.

15. A sound wave has frequency of 4 kHz and its wavelength is 35 cm, then how much time will it take to travel a distance of 3 km?

(a) 21.4 sec (b) 2.14 sec
(c) 12.4 sec (d) 1.25 sec

Chhattisgarh P.C.S. (Pre) 2021

Ans. (b)

The relation between wavelength (λ), frequency (n) and velocity (v) can be given as

$$v = n \times \lambda$$

Given that,

Frequency (n) = 4 kHz = 4×10^3 Hz ; Wavelength (λ) = 35 cm = 0.35 m; Distance (D) = 3 km = 3×10^3 m

Hence, Velocity of the sound wave (v) = $n \times \lambda$

$$= 4 \times 10^3 \times 0.35$$
$$= 1.4 \times 10^3 \text{ m/sec}$$

Now, Time (T) = $\frac{\text{Distance (D)}}{\text{Velocity } (v)}$

$$= \frac{3\times10^3}{1.4\times10^3} = 2.14 \text{ sec}$$

Thus, the given sound wave will take 2.14 sec to travel a distance of 3 km.

16. One important characteristic of sound is 'Pitch', which depends upon :

(a) Intensity (b) Frequency
(c) Quality (d) Amplitude

U.P. P.C.S. (Mains) 2017

Ans. (b)

The sensation of frequency is commonly referred to as the pitch of the sound. A high pitch sound corresponds to a high-frequency sound wave and a low pitch sound corresponds to a low-frequency sound wave. High pitch means very rapid oscillation and 'low' pitch corresponds to slower oscillation.

17. Shrillness of sound is determined by :

(a) velocity of sound
(b) amplitude of sound
(c) wavelength of sound
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

It is true that shrillness of sound is determined by its frequency but frequency and wavelength are inversely proportionate. So, it is also true to say that shrillness of sound depends on its wavelength. For example, shorter the wavelength means higher the frequency and more shrill sound. Since frequency and wavelength are inseparably interlinked (velocity of sound = wavelength × frequency and velocity of sound is constant in a medium), the correct answer is option (c).

18. Two astronauts cannot hear each other on the Moon's surface, because :

(a) Their ears have stopped working on the Moon.
(b) No atmosphere on the Moon.
(c) They wear special space suits on the Moon.
(d) The sound travels much more slowly on Moon.

U.P.U.D.A./L.D.A. (Pre) 2002

Ans. (b)

In order for sound to propagate from one place to another, it requires a medium or a fluid to move through. The air on the Earth allows sound waves to move from one point to another. However, there is vacuum on the surface of Moon. Thus, there is no sound on the Moon.

19. An astronaut cannot hear his companion at the surface of the Moon because :

(a) Produced frequencies are above the audio frequency.
(b) Temperature is too low during night and too high during day.
(c) There is no medium for sound propagation.
(d) There are many craters on the surface of the moon.

U.P.P.C.S. (Mains) 2013

Ans. (c)

See the explanation of above question.

20. Consider the following statements :

1. A flute of smaller length produces waves of lower frequency.

2. Sound travels in rocks in the form of longitudinal elastic waves only.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2007

Ans. (d)

A flute of smaller length produces the waves of lower wavelength and wavelength of any wave is inversely proportional to its frequency, thus a flute of smaller length produces waves of higher frequency. Therefore, statement 1 is not correct. Sound is transmitted through gases, plasma and liquids as longitudinal waves. Through solids (eg. in rocks), however, it can be transmitted as both longitudinal waves and transverse waves. Thus statement 2 is also incorrect.

21. Consider the following statements :

1. In a seismograph, P waves are recorded earlier than S waves.

2. In P waves, the individual particles vibrate to and fro in the direction of wave propagation whereas in S waves, the particles vibrate up and down at right angles to the direction of wave propagation.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2023

Ans. (c)

Seismic waves are vibrations in the Earth that transmit energy and occur during seismic activity such as earthquakes, volcanic eruptions, and even man-made explosions. There are two types of seismic waves, primary waves and secondary waves. Primary waves, also known as P waves or pressure waves, are longitudinal compression waves similar to the motion of a slinky. Thus, in P waves, the individual particles vibrate to and fro in the direction of wave propagation. P waves can travel through liquids, solids and gases. Secondary waves, or S waves which can only travel through solids,
are slower than P waves and therefore, in a seismograph, P waves are recorded earlier than S waves. The motion of secondary waves is perpendicular to the direction of the wave travel, similar to the motion of vigorously shaking a rope. Thus, in S waves, the particles vibrate up and down at right angles to the direction of wave propagation. Hence, both statements are correct.

22. Consider the following statements about ultrasonic waves :

1. They can destroy insects.
2. They can clean clothes by removing dust.
3. They can be used to treat diseases.
4. They can control automatic doors.

Of the above statements

(a) 1 and 2 are correct (b) 3 and 4 are correct
(c) 1, 2 and 3 are correct (d) All are correct

U.P.P.C.S. (Mains) 2004

Ans. (d)

Sound waves are divided into three categories that cover different frequency range; Audible waves (20 Htz – 20,000 Htz), Infrasonic waves (< 20 Htz) and Ultrasonic waves (> 20,000 Htz). Ultrasonic waves are used to destroy insects, clean clothes by removing dust, treat diseases, control automatic doors, detection of aircraft and submarine, determination of depth of sea etc.

23. Ultrasonics are sound waves of frequency :

(a) Greater than 20,000 Hz
(b) Less than 10,000 Hz
(c) Equal to 1000 Hz
(d) None of these

U.P.P.C.S.(Pre) 2012

Ans. (a)

See the explanation of above question.

24. What is the audible range (hearing range) of humans?

(a) 20 Hz - 20000 Hz (b) 80 Hz - 100 Hz
(c) 2 lac Hz - 4 lac Hz (d) 0 Hz - 20 Hz

M.P. P.C.S. (Pre) 2018

Ans. (a)

The sound that can be audible to humans has a frequency ranging from 20 Hz to 20,000 Hz.

25. What is/are not true among the following?

(i) Audible range of sound for human beings is approximately 20 Hz to 10 kHz.
(ii) Sound waves with frequencies higher than 10 kHz are called ultrasound.
(iii) In earthquake, ultrasounds are produced before main shock waves.

(a) (i) and (ii) (b) (i) and (iii)
(c) (i), (ii) and (iii) (d) Only (iii)

Chhattisgarh P.C.S. (Pre) 2022

Ans. (c)

Audible range of sound for human beings is approximately 20 Hz to 20 kHz (20,000 Hz). Sound waves with frequencies higher than 20 kHz (20,000 Hz) are called ultrasonic waves (ultrasound). In earthquake, infrasounds (less than 20 Hz) are produced before main shock waves. Hence, all three statements are incorrect.

26. A biotechnique in which ultrasonic sound is used :

(a) Sonography (b) E. C. G.
(c) E. E. G. (d) X-ray

R.A.S./R.T.S.(Pre) 2012

Ans. (a)

Sonography or ultrasonography is an important mean of clinical diagnosis. It is a diagnostic imaging technique based on the application of ultrasound (ultrasonic waves). It is widely used in the field of medical science. It is mainly used to provide a variety of information about the health of the mother during pregnancy, and the health and development of embryo or foetus.

27. **Which of the following rays/waves are used to know the growth of fetus in the womb?**
(a) X-rays (b) Microwaves
(c) Ultrasonic waves (d) Ultraviolet rays
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (c)

See the explanation of above question.

28. **Bats can fly during dark nights and also prey. This is because :**
(a) The pupil of their eyes is large.
(b) Their night vision is very good.
(c) Every bird can do this.
(d) They produce ultrasonic waves and are guided by them.

U.P.P.C.S. (Mains) 2005

Ans. (d)

Bats are a fascinating group of animals. They are one of the few mammals that can use ultrasonic sound to navigate. As they fly, make an ultrasonic (shouting) sound. The returning echoes give the bats information about anything that is ahead of them, including the speed and size of an insect and which way it is going. This system of finding prey is called echolocation- locating things by their echoes.

29. **Consider the following statements :**
1. The heart vibrates at infrasonic.
2. The speed of sound is more in gas than in liquid and solid.
3. Mach number is used to describe the speed of sound.
4. Ultrasonic sound has frequency more than 20,000 Hz .
Of these statements :
(a) 1, 2 and 3 are correct
(b) 2, 3 and 4 are correct
(c) 1, 3 and 4 are correct
(d) 1, 2 and 4 are correct

U.P.P.C.S. (Mains) 2002

Ans. (c)

The heart vibrates at infrasonic sound. We can hear them with the help of stethoscope. The speed of sound in solid is maximum. Mach number is a quantity that defines how quickly an object travels with respect to the speed of sound. The Mach number (M) is simply the ratio of the object's velocity (V) divided by the speed of sound at the altitude (a). Ultrasonic is the name given to sound waves that have frequencies greater than 20,000 Hz. Therefore, statement 1, 3 and 4 are correct.

30. **In Stethoscope, the sound of the patient's heartbeat reaches the doctor's ears by :**
(a) Multiple diffraction of sound
(b) Multiple reflection of sound
(c) Polarisation of sound
(d) Multiple refraction of sound

R.A.S./ R.T.S. (Pre) 2021

Ans. (b)

Stethoscopes work by the principle of multiple reflection of sound. When a doctor or nurse places a stethoscope diaphragm on a patient's chest, sound waves traveling through the patient's body cause the flat surface of the diaphragm to vibrate and because the vibrating object is attached to a tube, the sound waves are channeled in a specific direction. Each wave bounces, or reflects, off the inside walls of the rubber tube, a process called multiple reflection. In this way, each wave, in succession, reaches the eartips, or rubber nubs on the ends of the device, and finally the listener's eardrums.

31. **Which one of the following is the effect of the flight of supersonic jet?**
(a) Air pollution
(b) Eye disease
(c) Depletion in ozone layer
(d) None of these

M.P. P.C.S. (Pre) 1993

Ans. (c)

A jet engine is a machine for turning fuel into thrust. The thrust is produced by action and reaction also known as Newton's third law of motion. The force (action) of the exhaust gases pushing backward produces an equal and opposite force (reaction) called thrust, that powers the vehicle forward. Those jet planes which are able to fly faster than the speed of sound are called supersonic jet planes. Jet engines are responsible for depletion of ozone. Flying at stratosphere height, they emits nitrogen oxide which has the potential to destroy significant quantities of ozone in the stratosphere.

32. **Assertion (A) : A jet aircraft moving at mach number equal to 1 travels faster at an altitude of 15 km than while moving at mach number equal to 1 near the sea level.**
Reason (R) : The velocity of sound depends on the temperature of the surrounding medium.

Codes :

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(c) (A) is true, but (R) is false.

(d) (A) is false, but (R) is true.

I.A.S. (Pre) 2007

Ans. (d)

Speed of supersonic bodies is indicated by the mach number. The mach number of an airplane is the ratio of the speed of a body to the speed of sound in a particular medium, usually the atmosphere of the Earth. As we know that the speed of sound increases with increase in temperature, the speed of sound at higher altitude (15 km above sea level) decreases due to decrease in temperature. Thus, assertion (A) is false. The velocity of a sound wave travelling through the air does not vary with the air pressure but depends on the temperature of the air. Thus, reason (R) is correct.

33. Decibel is used to measure :

(a) Haemoglobin in blood
(b) Sugar in urine
(c) Sound in atmosphere
(d) Particles in air

M.P.P.C.S. (Pre) 2005

Ans. (c)

'Decibel' (dB) is used to measure the intensity of sound in atmosphere.

34. Decibel unit is used to measure :

(a) Light intensity
(b) Sound intensity
(c) Magnitude of Earthquake
(d) None of the above

U.P.P.C.S. (Mains) 2011

Ans. (b)

See the explanation of above question.

35. Decibel unit is used to measure the :

(a) Speed of light (b) Intensity of heat
(c) Intensity of sound (d) Radioactive frequency

Uttarakhand Lower Sub. (Pre) 2010

Ans. (c)

See the explanation of above question.

36. What is the decibel level of sound produced by two persons in conversation?

(a) About 5 Decibel (b) About 10 Decibel
(c) About 30 Decibel (d) About 100 Decibel

R.A.S./R.T.S.(Pre) 2003

Ans. (c)

The intensity of sound is measured in decibel.

Source of sound	Intensity (In decibel)
Whisper	15–20
Normal Conversation	30–60
Anger Conversation	70–80
Truck-Motorcycle	90–95
Instrument factory or machine shop	100–110
Orchestra	110–120
Jet Plane	140–150

37. A noise level of 100 decibel would correspond to :

(a) Just audible sound
(b) Ordinary conversation
(c) Sound from a noisy street
(d) Noise from a machine shop

I.A.S. (Pre) 2000

Ans. (d)

See the explanation of above question.

38. Which of the following represents the decibel level of rustling of tree leaves in normal circumstance?

(a) 10 dB (b) 20 dB
(c) 60 dB (d) 100 dB

U.P. P.C.S. (Pre) 2018

Ans. (b)

In normal circumstances, the decibel level of the rustling of tree leaves is about 20 decibel. A decibel (dB) is 1/10 of a bel, which is a unit of sound intensity named in the honour of Alexander Graham Bell.

39. As per the WHO, the safe noise level for a city is –

(a) 45 dB (b) 50 dB
(c) 55 dB (d) 60 dB

U.P.P.C.S. (Mains) 2010

Ans. (a)

As per the WHO the safe noise level for a city is 45 decibel (dB).

40. In the context of permissible noise levels match List-I with List-II and select the answer from the code given below the lists.

List-I (Area)	List-II (Permissible Noise Level)
A. Residential area	**1. 50 dB**
B. Silent zone	**2. 55 dB**

C. Industrial area **3. 65 dB**

D. Commercial area **4. 70 dB**

Code :

	A	B	C	D
(a)	3	4	2	1
(b)	1	2	3	4
(c)	2	1	3	4
(d)	2	1	4	3

U.P. P.C.S. (Pre) 2022

Ans. (d)

As per the Noise Pollution (Regulation and Control) Rules, 2000 of India, ambient air quality standards in respect of noise (permissible noise levels) in given areas are as follows :

Area	Permissible Noise Level (dB)	
	Day Time	Night Time
Residential area	55	45
Silent zone	50	40
Industrial area	75	70
Commercial area	65	55

41. The tolerable limit of noise for human being is around:

(a) 45 decibel (b) 85 decibel

(c) 125 decibel (d) 155 decibel

R.A.S./R.T.S. (Pre) 1993

Ans. (b)

The intensity of sound in normal conversation is 30-60 decibel. The sound of 50 decibel is enough to wake up a sleeping person. To remain continuously in the sound of 80 decibel proves to be harmful. 90 decibel is the maximum limit to tolerate any noise by human being.

42. Sound above what level (in decibels) is considered hazardous noise pollution?

(a) 30 dB (b) 100 dB

(c) 80 dB (d) 120 dB

U.P.P.C.S.(Pre) 2013

U.P.P.C.S. (Mains) 2008

Ans. (c)

Sound above 80 decibel is considered hazardous noise pollution. Continued exposure to high levels of noise results in fatigue, hearing loss or even total loss of hearing, changes in blood circulation, changes in breathing patterns etc.

43. Which one of the following units is used for measurement of noise pollution?

(a) Nanometer (b) Decibel

(c) Hertz (d) None of the above

U.P. P.C.S. (Mains) 2017

Ans. (b)

See the explanation of above question.

44. The optimum sound level for human beings is–

(a) 90 dB (b) 60 dB

(c) 120 dB (d) 100 dB

R.A.S./R.T.S.(Pre) 2012

Ans. (b)

The average sound level in normal conversation is 60 decibel, It is appropriate for human ears. The sound higher than 80 decibel can cause damage to the ear cells.

45. Loudness of sound is measured in terms of following :

(a) Frequency (b) Amplitude

(c) Velocity (d) Wavelength

U.P. R.O./A.R.O. (Mains) 2017

Ans. (b)

A sound wave's amplitude is the change in pressure as the sound wave passes by. If you decrease the amplitude, you are making the sound softer. The amplitude of a wave is related to the amount of energy it carries. A high amplitude wave carries a large amount of energy, a low amplitude wave carries a small amount of energy. The average amount of energy passing through a unit area per unit of time in a specified direction is called the intensity of the wave. As the amplitude of the sound wave increases, the intensity of sound increases. Sounds with higher intensity are perceived to be louder. Relative sound intensities are often given in units named decibel (dB).

46. A worker was working on the rail track. A boy at a distance holds his ear near the rail track. This boy was able to hear sound of the blow of workman twice. It is because of :

(a) The speed of sound is greater in steel than in air.

(b) The speed of sound is greater in air than in steel.

(c) Part of the sound wave is reflected between the rail tracks.

(d) His ears are at different distance from the source.

R.A.S./R.T.S. (Pre) 1994-95

Ans. (a)

The speed of sound is greater in steel than in air, as sound reaches first by rail track and then by air. Therefore the boy hears the sound twice.

47. Which of the following statements is/are true in relation to sound waves?

Statement I : Speed of sound decreases when we go from solid to gaseous state

Statement II : In any medium as we decrease the temperature, the speed of sound increases

Statement III : Speed of sound is less in aluminium medium than in glass medium

(a) Statement I, II and III all are true.
(b) Only Statement I and II are true.
(c) Only Statement I is true.
(d) Only Statement II and III are true.

Chhattisgarh P.C.S. (Pre) 2020

Ans. (c)

The Speed of sound depends upon the nature of medium through which it propagates. Speed of sound is maximum in solids while minimum in gases. Speed of sound is less in glass medium than in aluminium medium. Increase in temperature results to increase in the speed of sound in any medium. Hence, among the given statement, only statement I is true.

48. When a sound wave goes from one medium to another, the quantity that remains unchanged is :

(a) Frequency (b) Amplitude
(c) Wavelength (d) Speed
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (a)

When a sound wave goes from one medium to another, its frequency usually remains the same (because it is like a driven oscillation and maintains the frequency of the original source) while its speed and wavelength are changed. The amplitude of sound is also changed as it decreases with distance from its source.

49. Before playing the orchestra in a musical concert, a sitarist tries to adjust the tension and pluck the string suitably. By doing so, he/she is adjusting :

(a) frequency of the sitar string with the frequency of other musical instruments
(b) amplitude of sound
(c) intensity of sound
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

String length, diameter, density and tension determine the frequency in which it will vibrate. Since other three are fixed, it is the tension that sitarist tries to adjust to tune its natural frequency. And while playing the Sitar, he changes the effective length of the string by touching the string at different points with his finger. This is how we get different melodious notes (of varying frequency). Amplitude/intensity of sound is not determined by the tension in the string since it depends on how the Sitarist is playing. Even with a perfectly taut string, he can produce low or very high sound. But he cannot vary its natural frequency and that's why he has to adjust the tension beforehand. So, the correct answer is (a).

50. When the same tone is played on a sitar and a flute, the sound produced can be distinguished from each other because of the difference in :

(a) pitch, loudness, and quality
(b) pitch and loudness
(c) quality only
(d) loudness only

I.A.S. (Pre) 1995

Ans. (c)

Musical sound can differ from each other with respect to the following three characteristics :

1. Loudness (intensity)
2. Pitch (shrillness)
3. quality (timber)

But quality (timber) is that characteristic of a musical sound which enables us to distinguish between two sounds even if they have the same pitch and loudness. Hence, when the same tone is played on a sitar and a flute, the sound produced can be distinguished from each other because of the difference in quality.

51. The basic units of sound are called–

(a) Morphemes (b) Phonemes
(c) Semanteme (d) Syntax

U.P. P.C.S. (Mains) 2017

Ans. (b)

Phonemes is the basic unit of sound (phonology). It is the smallest unit of sound that may cause a change of meaning within a language, but that does not have meaning by itself.

52. Assertion (A) : Reverberation mainly feels in large churches and in other large buildings.

Reason (R) : The walls, roof and ground may cause multiple sound reflections.

Code :

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(c) (A) is true, but (R) is false.

(d) (A) is false, but (R) is true.

U.P. Lower Sub. (Spl.) (Pre) 2003

U.P. Lower Sub. (Spl.) (Pre) 2002

Ans. (a)

Reverberation is a result of multiple reflections. A sound wave in an enclosed or semi-enclosed environment (like large church or in other big building) will be broken up as it is bounced back and forth among the reflecting surfaces. Reverberation is an effect which is multiplicity of echoes whose speed of repetition is to be quick for them and to be perceived as separate from one another.

53. Consider the following statements:

1. A widely used musical scale called diatonic scale has seven frequencies.

2. The frequency of the tone Sa is 256 Hz and that of Ni is 512 Hz.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2008

Ans. (d)

The most common musical scale is the diatonic scale. It consists of a series of eight notes or tones, and the ratio between the last and the first note is 2 : 1. It is therefore called an octave. The major diatonic scale consists of notes with frequencies in the ratio 1, 9/8, 5/4, 4/3, 3/2, 5/3, 15/8 and 2 relative to a key note. The series of notes is defined as:

C	D	E	F	G	A	B	C_1
(Sa)	(Re)	(Ga)	(Ma)	(Pa)	(Dha)	(Ni)	(Sa_1)

Here, the higher frequency Sa(Sa_1) is called the octave of the lower Sa. If the frequency of first note (Sa) is set as 256 Hz, then that of note Ni is 480 Hz (256 × 15/8) and the frequency of octave Sa_1 is 512 Hz (256 × 2). The above explanation shows that both the given statements are incorrect.

54. Television signals cannot be received beyond a certain distance because :

(a) Signals are weak

(b) Antenna is weak

(c) Air absorbs signals

(d) The surface of the Earth is curved

U.P.P.S.C. (GIC) 2010

U.P.P.C.S. (Pre) 1994

Ans. (d)

Television signals cannot be received beyond a certain distance because the surface of the Earth is curved, due to this the signals moves further without hitting the Earth's surface.

55. When T.V. is switched on :

(a) Audio and video both start simultaneously

(b) Audio is heard immediately but video starts later because video needs some warm up time

(c) Video starts immediately but audio is heard later because sound travels at a lesser speed than light

(d) It depends on the T.V. brand

U.P.P.C.S. (Pre) 2007

Ans. (a)

When T.V. is switched on, audio and video start simultaneously. In old models of television, audio was heard immediately but the video starts as it needs some warm up time. But in modern televisions, audio synchronizer is used to correct this sync error.

56. In television broadcast, the picture signals are transmitted by –

(a) Amplitude modulation

(b) Frequency modulation

(c) Phase modulation

(d) Angle modulation

R.A.S./R.T.S.(Pre) 2012

Ans. (a)

In a television broadcast, the picture signals are transmitted by amplitude modulation and audio signals are transmitted by frequency modulation.

57. The technique used to transmit audio signals in television broadcasts is :

(a) Amplitude Modulation

(b) Frequency Modulation

(c) Pulse Code Modulation

(d) Time Division Multiplexing

U.P.P.C.S. (Mains) 2007

I.A.S. (Pre) 1995

U.P.U.D.A./L.D.A. (Spl.) (Pre) 2010

Ans. (b)

Frequency modulation is used to transmit audio signals in television broadcasts. Frequency modulated signals have larger bandwidth so that FM signals on the adjacent bands have neither noise nor interference issue.